

MSK IMCQ 05

Histology & Biochemistry Key

Biochemistry Questions

A 47-year-old man comes to the physician because of a 2-week history of left hip pain. He has a history of hypertension, type 2 diabetes mellitus, and stage-4 chronic kidney disease. Physical examination shows pain on external rotation of the hip. An x-ray of the hip shows subperiosteal resorption. Which of the following laboratory findings is most likely in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V

	Serum PTH	Serum calcium
I	Elevated	Elevated
II	Elevated	Decreased
III	Decreased	Decreased
IV	Decreased	Elevated
V	Normal	Normal

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A 37-year-old woman comes to the physician for a follow-up examination 2 weeks after she underwent total thyroidectomy. She says that after being discharged she feels fatigued, lethargic and has an increased sensitivity to feeling cold. Physical examination shows spasmic movements of her face when the facial nerve is tapped. A comprehensive metabolic panel is performed. Which of the following laboratory findings are most likely in this patient?

		Parathyroid hormone (PTH)	Serum Calcium	Serum Phosphorus
A.	Option 1	Increased	Increased	Decreased
B.	Option 2	Increased	Decreased	Decreased
C.	Option 3	Decreased	Decreased	Increased
D.	Option 4	Decreased	Increased	Increased
E.	Option 5	Increased	Increased	Increased
F.	Option 6	Decreased	Decreased	Decreased

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SOM.MK.I.BPM1.2.MSK.1.BCHM.0211
Describe the associations among intestinal absorption, renal elimination, bone stores, and the function of calcitriol and parathyroid hormone in regulating serum calcium and phosphate

		Parathyroid hormone (PTH)	Serum Calcium	Serum Phosphorus
A.	Option 1	Increased	Increased	Decreased
B.	Option 2	Increased	Decreased	Decreased
 C.	Option 3	Decreased	Decreased	Increased
D.	Option 4	Decreased	Increased	Increased
E.	Option 5	Increased	Increased	Increased
F.	Option 6	Decreased	Decreased	Decreased

A 32-year-old man comes to the physician because of a 2-month history of persistent left leg pain and muscle weakness. He has a life-long history of cystic fibrosis and a 2-year history of cirrhosis of the liver. Serum studies show calcium concentration is 7.8 mg/dL (8.4-10.2 mg/dL), alkaline phosphatase (ALP) is 240 U/L (25-100 U/L), and a low 25-hydroxyvitamin D concentration. Which of the following would most likely be elevated in this patient?

- A. Calbindin expression in intestine
- B. 1,25-dihydroxyvitamin D
- C. Serum parathyroid hormone
- D. Serum phosphorous
- E. Calcitonin

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A 56-year-old woman comes to the physician because of a 6-week history of constant fatigue, anorexia, polyuria, and constipation. Abdominal ultrasonography shows renal calculi. Laboratory studies show serum calcium is 11.5 mg/dL (8.4 - 10.2 mg/dL), serum phosphorus is 2.8 mg/dL (3.0 - 4.5 mg/dL), and parathyroid hormone concentration is 800 pg/mL (Normal: 10-60 pg/mL). Which of the following sets of pathophysiologic changes are most likely in this patient?

		Bone resorption	Renal Calcium reabsorption	Renal Calcitriol synthesis
A.	I	Elevated	Reduced	Elevated
B.	II	Elevated	Elevated	Reduced
C.	III	Elevated	Elevated	Elevated
D.	IV	Reduced	Reduced	Reduced
E.	V	Reduced	Reduced	Elevated
F.	VI	Reduced	Elevated	Elevated

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SOM.MK.I.BPM1.2.MSK.1.BCHM.0209 Describe hypocalcemia and hypercalcemia as well as their causes, manifestations, diagnosis and treatment

		Bone resorption	Renal Calcium reabsorption	Renal Calcitriol synthesis
A.	I	Elevated	Reduced	Elevated
B.	II	Elevated	Elevated	Reduced
✓ C.	III	Elevated	Elevated	Elevated
D.	IV	Reduced	Reduced	Reduced
E.	V	Reduced	Reduced	Elevated
F.	VI	Reduced	Elevated	Elevated

A 7-year-old girl is brought to the physician by her mother because of a 3-month history of bowing of her legs and hair loss. Laboratory studies show hypocalcemia, hypophosphatemia, and elevated plasma calcitriol concentration. Which of the following is most likely in this patient?

- A. Reduced exposure to sunlight
- B. Excessive supplements of vitamin D
- C. Vitamin D resistant rickets
- D. Hypoparathyroidism
- E. Excessive calcitonin secretion
- F. Chronic liver disease
- G. Chronic kidney disease

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SOM.MK.I.BPM1.2.MSK.1.BCHM.0214 Correlate hereditary resistance to vitamin D (vitamin D resistant rickets) to the biochemical mechanisms

A 72-year-old woman is admitted to the hospital after she fell and broke her left ankle. She has a 2-year history of chronic renal failure requiring hemodialysis. Results of laboratory studies are shown. Which of the following is most likely reduced in this patient?

Serum parameter	Patient lab values	Normal Range
Calcium	7.8 mg/dL	8.4 – 10.2 mg/dL
Alanine amino transferase (ALT)	39 U/L	10 – 40 U/L
Creatinine	3.6 mg/dL	0.6 – 1.2 mg/dL
Phosphorus	5.0 mg/dL	3.0 – 4.5 mg/dL

- A. Formation of 7-dehydrocholesterol
- B. Formation of 25-hydroxycholecalciferol
- C. Bone demineralization
- D. Gene expression of calbindin
- E. Parathyroid hormone secretion

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Histology Questions

A 68-year-old woman is brought to the emergency department because of sudden onset lower back pain following a minor fall while attempting to lift a flower-pot. Her blood pressure is 132/78 mm Hg, pulse is 78/min, respirations are 16/min and temperature is 36.8°C (98.24°F). Physical examination shows thoracolumbar kyphosis, difficulty standing upright and persistent discomfort in the lower back when standing, walking or bending, as well as pain on palpation of the L3-L4 lumbar region. An X-ray of the spine shows compression fractures at the L3 and L4. Bone density scan shows low bone density. Which of the following will most likely be observed on a biopsy of the bone?

- A. Sparse, large, eosinophilic staining, multi-nucleated cells in lacunae
- B. Abundant cells containing PAS positive staining granules
- C. Highly active basophilic staining cells in the periosteum
- D. Highly active stellate-shaped cells with normal canicular activity
- E. Abundant, large, eosinophilic staining, multi-nucleated cells in resorption lacunae

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A 24-year-old man comes to the physician because of a 3-month history of progressively worsening, dull pain in the left femur when walking or climbing stairs. Physical examination shows a mild swelling over the mid left femur, no erythema and normal range of motion. An X-ray of the femur shows a radiolucent lesion in the diaphysis of the left femur with surrounding sclerosis. Examination of a biopsy specimen from the lesion shows random orientation of the collagen fibers. Which of the following is most likely an additional observation in the biopsy?

- A. Well-organized Volkmann's canals
- B. Abundance of concentric lamellae
- C. Increased osteoblastic and osteoclastic activity
- D. Well-developed, normal appearing osteons
- E. Regular distribution of the osteocytes

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A 42-year-old woman comes to the physician because of a 6-month history of progressive tiredness and shortness of breath. She has a history of heavy menstrual bleeding and iron-deficiency anemia, previously treated with oral iron. Her blood pressure is 118/76 mmHg, pulse is 92/min, respirations are 18/min and temperature is 36.7°C (98.06°F). Physical examination shows pale conjunctivae and no lymphadenopathy, hepatosplenomegaly or bone tenderness. Laboratory studies of the blood is shown in the attached table. A peripheral smear confirms microcytic, hypochromic RBCs. Endoscopic studies show destruction of the microvilli of the small intestines, thus affecting absorption. Which of the following features of bone tissue is most likely to be affected by this patient’s condition?

- A. Remodeling triggered by mechanical strain
- B. Proper bone formation and mineralization
- C. Slow rate of initiation of osteolysis
- D. Inability for osteoprogenitor cells to differentiate to osteoblasts
- E. Inability to stimulate epiphyseal cartilage growth

Laboratory Parameter	Patient Value	Normal Value
Hemoglobin	8.2g/dL	12 – 15 g/dL
MCV	72 fL	80 – 100 fL
Ferritin	9 ng/mL	13 – 150 ng/mL
Reticulocyte count	3%	0.5% - 2%

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A group of researchers is studying the effects of growth hormone on height. The study involves observation of the epiphyseal growth plate in vitro. Which of the epiphyseal zones in the attached photomicrograph is most likely influenced by the hormone being studied?

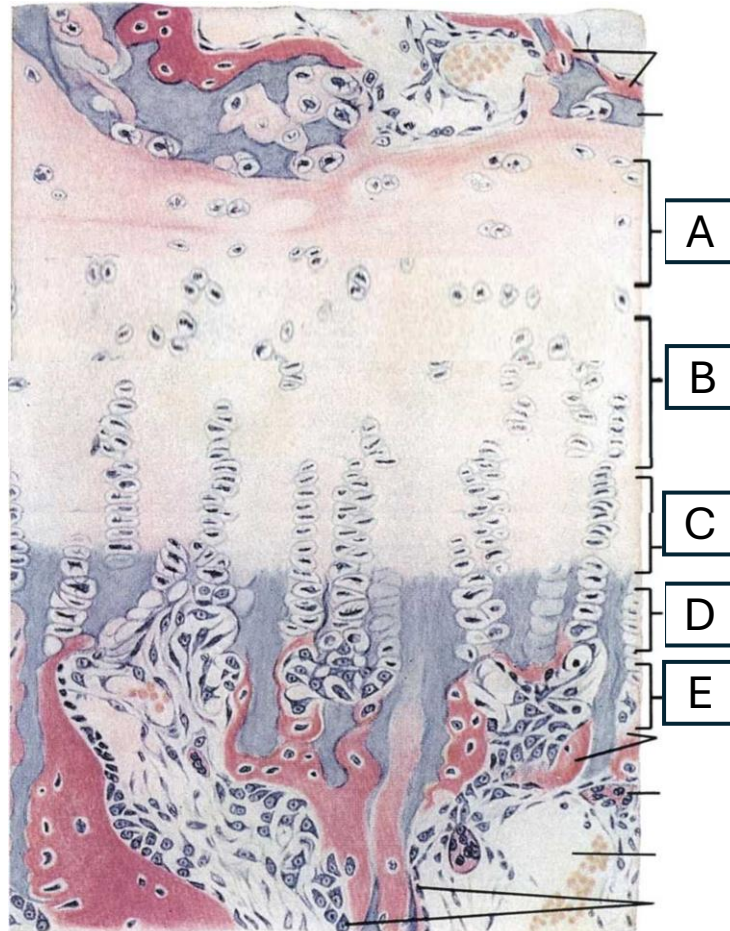
A. A

B. B

C. C

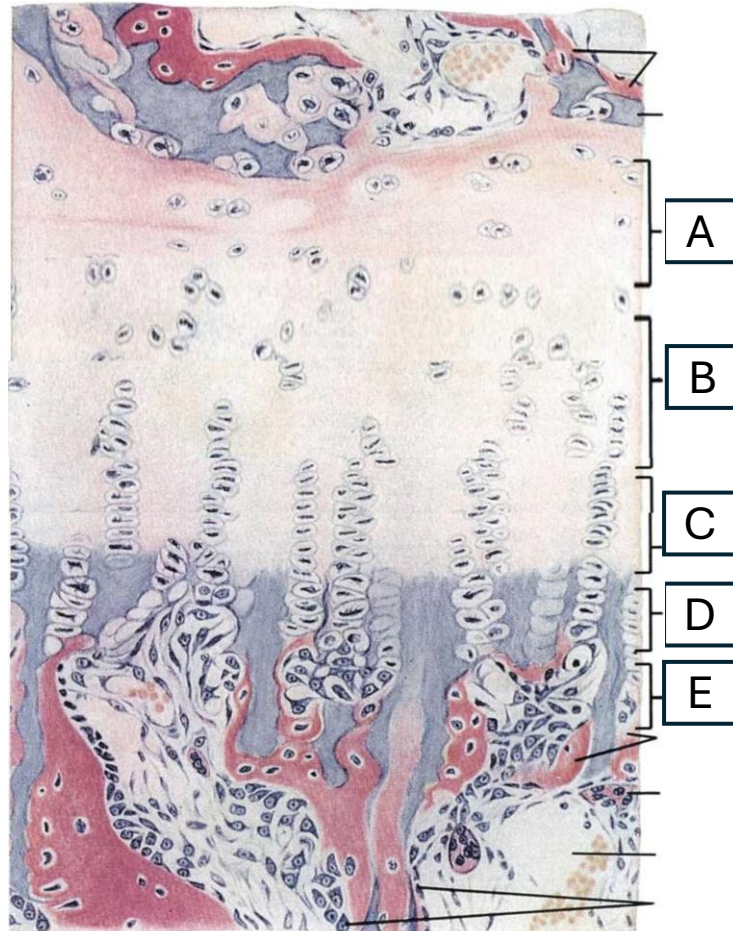
D. D

E. E



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- A. A
- B. **B**
- C. C
- D. D
- E. E



A 12-year-old girl is brought to the emergency department by her father because of a 1-hour history of pain in the wrist following a fall during a skateboarding accident 1 hour ago. Physical examination shows a swollen, tender, red right wrist with no obvious deformities. An X-ray of the wrist shows no signs of fracture and the well-defined radiolucent line on the distal radius remained intact (see attached). The physician explains to the father that the radiolucent line is the growth (epiphyseal) plate, and this will eventually fuse as the bone matures. Which of the following is most likely a function of this region of the bone?

- A. Responsible for the production of the matrix of bone
- B. Serves as a calcium reserve for the body
- C. Contributes to the longitudinal growth of bones
- D. Involved in the production of the granulation tissue during the bone repair process
- E. Aids in the mineralization process of bone tissue



https://www.rch.org.au/uploadedImages/Main/Content/clinicalguide/Fracture-Distal-radial-physeal-Figure-2_1396963-salter-harris1_AP.jpg

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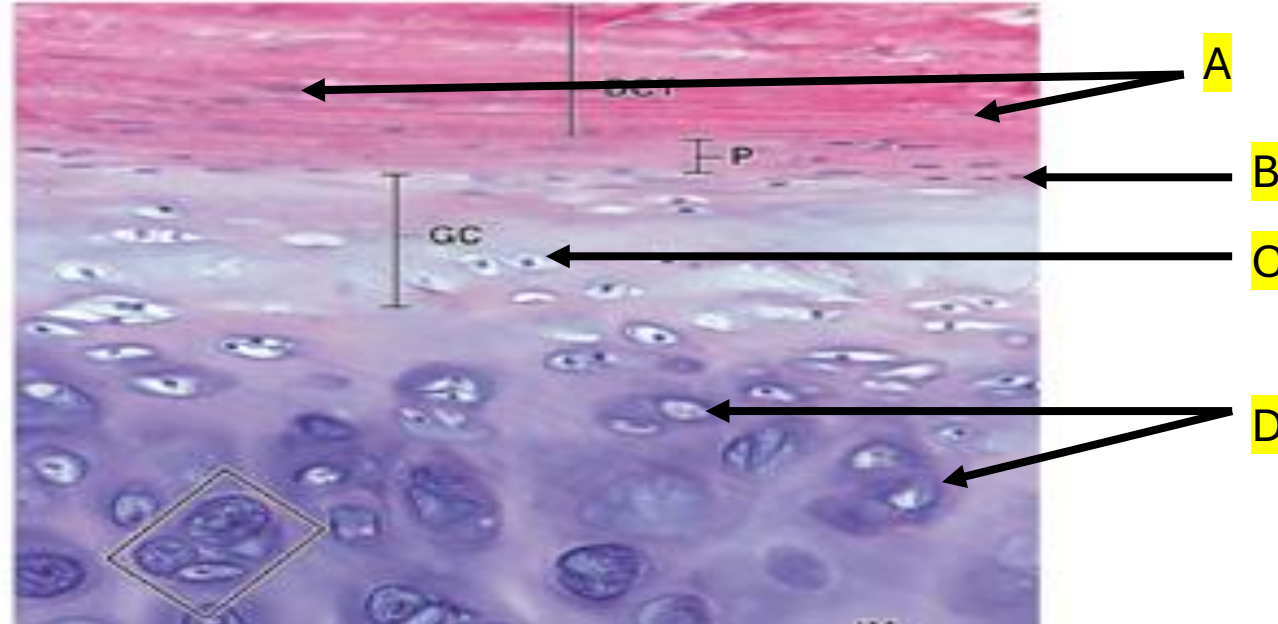
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A 55-year-old man comes to the physician with a 3-month history of worsening shortness of breath especially when lying flat or on exertion. Physical examination shows occasional wheezing only. A chest x-ray shows narrowing of the upper airway at the level of the trachea. A CT-scan of the neck and chest show thickening of the tracheal wall. Microscopic examination of a biopsy specimen of the tracheal wall shows hyperplasia of cells arranged in isogenous groups within lacunae. These cells are individually surrounded by a distinct pericellular matrix. Isogenous groups are enclosed by a more fibrous territorial matrix and are separated by the pale-staining interterritorial extracellular matrix. Which of the following cell types in the attached photomicrograph is most likely affected?

- A. A
- B. B
- C. C
- D. D



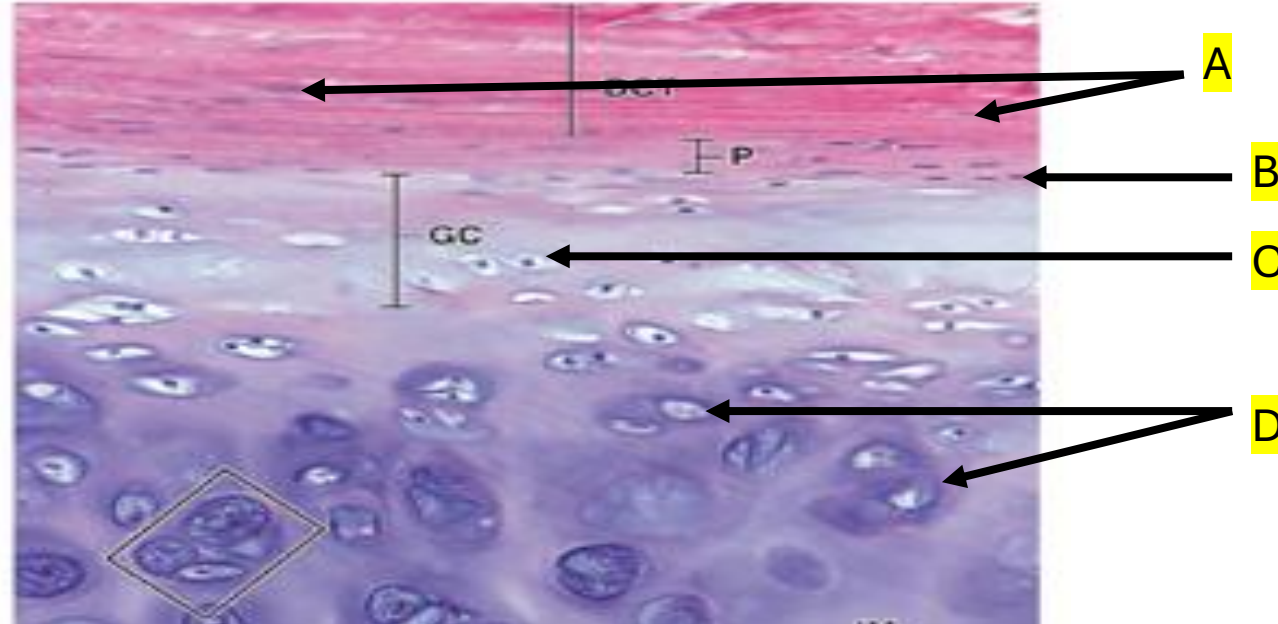
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A. A

B. B

C. C

D. D



A 12-year-old girl is brought to the physician by her parents because of a 2-year history of joint pain and stiffness in her knees and ankles. Physical examination shows decreased range of motion both passively and actively in the knee and ankle joint bilaterally. An x-ray of her knee and ankle joints show irregular epiphyseal plate development and shortened long bones. Genetic testing shows a mutation in the gene that codes for aggrecan. Which of the following best classifies the affected extracellular matrix component?

- A. Proteoglycan
- B. Multiadhesive glycoprotein
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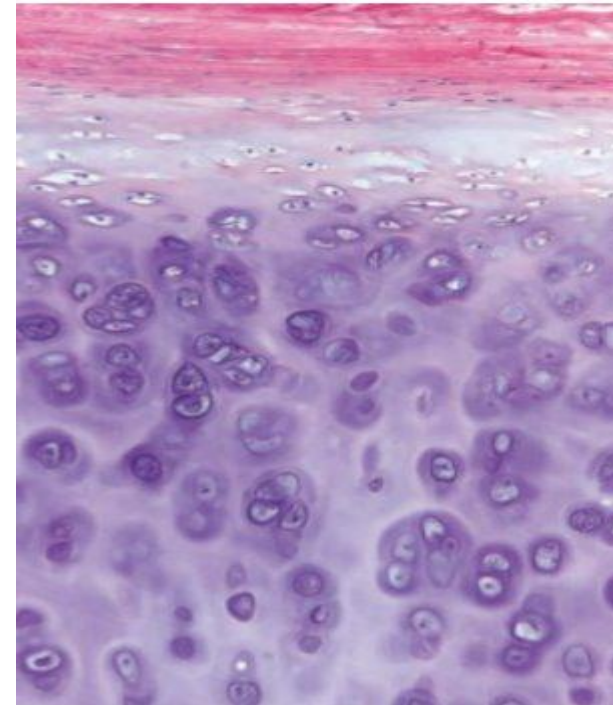
- A. Cartilage lengthening which occurs by mitotic division of chondrocytes is the only type of growth observed
- B. Absence of chondrocytic mitotic activities and isogenous groups
- C. Chondroblasts depositing new ECM peripherally and chondrocytes in isogenous groups secreting new ECM centrally
- D. Transient halting of mitotic activity of the chondrocytes along with loss of chondrogenic cells, resulting in both shortening and shrinkage of the cartilage
- E. Cartilage widening which occurs from the perichondrium is the only type of growth seen

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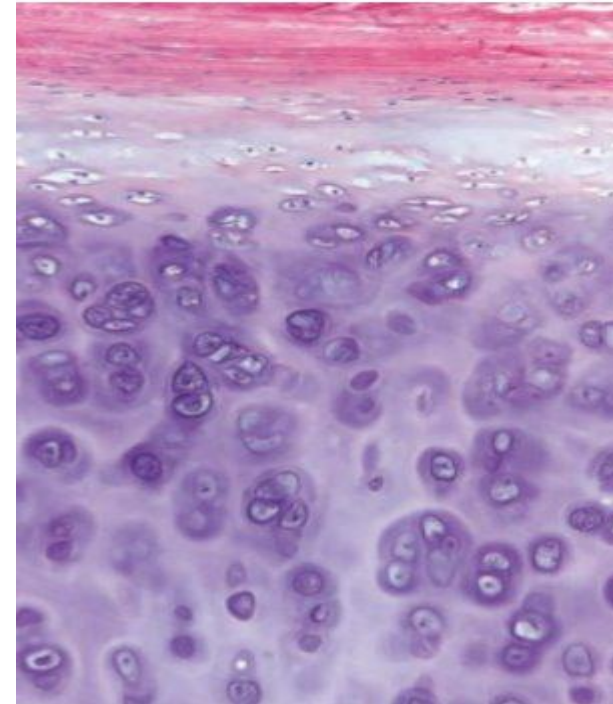
A 7-year-old boy is brought to the physician by his parents because of a 1-year history of gradual bulging outward of the tip of his nose. There is no history of trauma or infection. Physical examination show a mass on the nasal septum (cartilaginous portion of the nose). Examination of a biopsy specimen of the mass show chondrocytes in lacunae, occasionally in isogenous groups, with invasion into the adjacent perichondrial layer. A normal photomicrograph of the tissue in which the mass was seen is attached for your reference. Which of the following best characterizes the type of tissue in which the mass was found?

- A. Lacks a perichondrium
- B. Elastic fibers and Type II collagen
- C. Type I and Type II collagen
- D. Fibroblasts and chondrocytes
- E. Undergoes interstitial growth only
- F. Type II collagen is the most abundant fiber type



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A 32-year-old woman comes to physician because of a 7-day history of persistent pain and swelling over the anterior aspect of her right knee, particularly when kneeling or ascending stairs. She has a history of a direct blow to the patella during a fall 2 weeks ago. Physical examination shows mild effusion in the knee joint and point tenderness over the patellar cartilage. MRI of the knee shows a focal defect in the articular cartilage surface of the patella, with preserved underlying bone and no obvious extension into subchondral bone.

The physician explains that unlike elastic cartilage or fibrocartilage, hyaline cartilage in joints **does not** have a perichondrium over the articulating surface, and therefore intrinsic repair capacity is extremely limited. Which of the following best describes a function of the absent structure in this type of tissue?

- A. Secretion of the fibers of the extracellular matrix
- B. Contributes to growth in lengthening of the cartilage
- C. Production of the ground substance of the extracellular matrix
- D. Contributes to the growth in width of the cartilage
- E. Provides the vascularity of the tissue

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